

Difference between HDLC and PPP

When it comes to information connect conventions, HDLC (High-Level Information Connect Control) and PPP (Point-to-Point Convention) are two broadly recognized guidelines. Whereas both conventions encourage the exchange of information between arranged gadgets, they contrast in a few crucial angles. This article points to shed light on the difference between HDLC and PPP, investigating their definitions, standardization, surrounding strategies, tending to components, mistake-taking care of capabilities, transaction highlights, verification back, association sorts, overhead contemplations, and applications.

What is HDLC?

High-level Data Link Control (HDLC) could be a synchronous bit-oriented information interface layer convention utilized for transmitting data over point-to-point and multipoint joins. It could be a broadly utilized convention that gives solid and proficient information transmission between organized gadgets.

HDLC operates in two primary modes –

Normal Response Mode (NRM) and Asynchronous Balanced Mode (ABM). In NRM, one gadget called the essential station, controls communication by starting and overseeing information exchange. ABM, on the other hand, permits both gadgets to act as breaks even, with either gadget starting communication or both gadgets having risen to duties.

HDLC frames comprise a header, data field, and a trailer. The header contains control data, such as the address of the goal station, control bits for stream control, and mistake discovery data. The data field carries the real information being transmitted. The trailer contains a cyclic repetition check (CRC) esteem for mistake discovery.

HDLC bolsters different transmission modes, counting full-duplex and half-duplex modes. It moreover gives components for blunder discovery and redress, stream control, and multiplexing different consistent channels over a single physical connection.

The HDLC convention has been broadly embraced in numerous organizing innovations, counting synchronous serial interfaces, Integrated Administrations Computerized Organize (ISDN), and X.25 systems. It serves as an establishment for other conventions such as Cisco's proprietary convention, Point-to-Point Convention (PPP), and subsidiaries like Outline Hand-off.

HDLC utilizes cyclic repetition check (CRC) for mistake discovery, which includes adding a checksum to the outline to confirm information keenness amid transmission. In any case, it does not give built-in blunder redress components.

In spite of the fact that HDLC was initially created for utilization in synchronous serial communication joins it has found far-reaching appropriation in different organized situations.

What is PPP?

Point-to-Point Protocol (PPP) is a Synonyms information interface layer convention utilized for building up a coordinate association between two arrange hubs, regularly over a serial interface. It gives a standard strategy for transmitting information bundles over different physical media, such as serial cables, phone lines, or fiber optic joins.

PPP offers a solid and effective way to set up and keep up a communication connection between two gadgets, permitting them to trade network-layer parcels. It underpins confirmation, blunder location, and multilink capabilities.

PPP employs a Link Control Protocol (LCP) to arrange and design the association parameters, such as verification strategies and organize conventions to be utilized

PPP too underpins different confirmation strategies, counting Secret word Confirmation Convention and Challenge Handshake Confirmation Convention, which guarantee secure and confirmed associations between arranged gadgets.

Differences between HDLC and PPP

The differences are in the following table –

Basis of Difference	HDLC	PPP
Definition	Bit-oriented synchronous information interface convention.	Byte-oriented non-concurrent information connects convention.

Standardization	It has been standardized by an international standards organization and is widely recognized and used across the world.	The standardization followed by the protocol is IETF and it is an open standard so can be used by people very easily.
Encapsulation	They can carry data from different network layer protocols within their frames	PPP have the ability to encapsulate different protocols.
Authentication	It does not provide a way to verify the identity of the devices communicating over the network.	These mechanisms allow the devices communicating over the network to verify each other's identity before establishing a connection.
Application	It is often used for transmitting data between devices within a local network.	PPP is commonly used for dial-up and PPPoE (Point-to-Point Protocol over Ethernet) connections.

Conclusion

In conclusion, HDLC and PPP are information interface conventions with shifting characteristics and functionalities. HDLC, being a bit-oriented synchronous convention, gives direct overhead. On the other hand, PPP, a byte-oriented no concurrent convention, is predominant in dial-up associations and underpins broad arrangement capabilities, verification instruments, and variable-length surrounding. Whereas both conventions offer mistake discovery utilizing CRC, they need built-in blunder redress highlights. By comprehending the incongruities between HDLC and PPP, organize experts can successfully select the suitable convention to guarantee proficient and secure information transmission in differing arrange situations.

PPP Protocol

The PPP stands for Point-to-Point protocol. It is the most commonly used protocol for point-to-point access. Suppose the user wants to access the internet from the home, the PPP protocol will be used.

It is a data link layer protocol that resides in the layer 2 of the [OSI model](#). It is used to encapsulate the layer 3 protocols and all the information available in the payload in order to be transmitted across the serial links. The PPP protocol can be used on synchronous link like ISDN as well as asynchronous link like dial-up. It is mainly used for the communication between the two devices.

It can be used over many types of physical networks such as serial cable, phone line, trunk line, cellular telephone, fiber optic links such as SONET. As the data link layer protocol is used to identify from where the transmission starts and ends, so ISP (Internet Service Provider) use the PPP protocol to provide the dial-up access to the [internet](#).

Services provided by PPP

- It defines the format of frames through which the transmission occurs.
- It defines the link establishment process. If user establishes a link with a server, then "how this link establishes" is done by the PPP protocol.
- It defines data exchange process, i.e., how data will be exchanged, the rate of the exchange.
- The main feature of the PPP protocol is the encapsulation. It defines how network layer data and information in the payload are encapsulated in the data link frame.
- It defines the authentication process between the two devices. The authentication between the two devices, handshaking and how the password will be exchanged between two devices are decided by the PPP protocol.

Services Not provided by the PPP protocol

- It does not support flow control mechanism.
- It has a very simple error control mechanism.
- As PPP provides point-to-point communication, so it lacks addressing mechanism to handle frames in multipoint configuration.

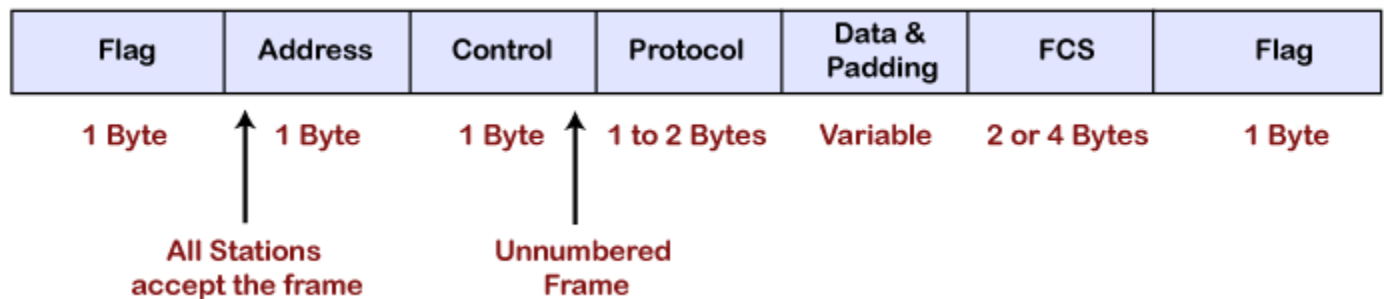
It is a byte-oriented protocol as it provides the frames as a collection of bytes or characters. It is a WAN (Wide Area Network) protocol as it runs over the [internet](#) link which means between two routers, internet is widely used.

PPP has two main uses which are given below:

- It is widely used in broadband communications having heavy loads and high speed. For example, an internet operates on heavy load and high speed.
- It is used to transmit the multiprotocol data between the two connected (point-to-point) computers. It is mainly used in point-to-point devices, for example, routers are point-to-point devices where PPP protocol is widely used as it is a WAN protocol not a simple LAN ethernet protocol.

Frame format of PPP protocol

The frame format of PPP protocol contains the following fields:

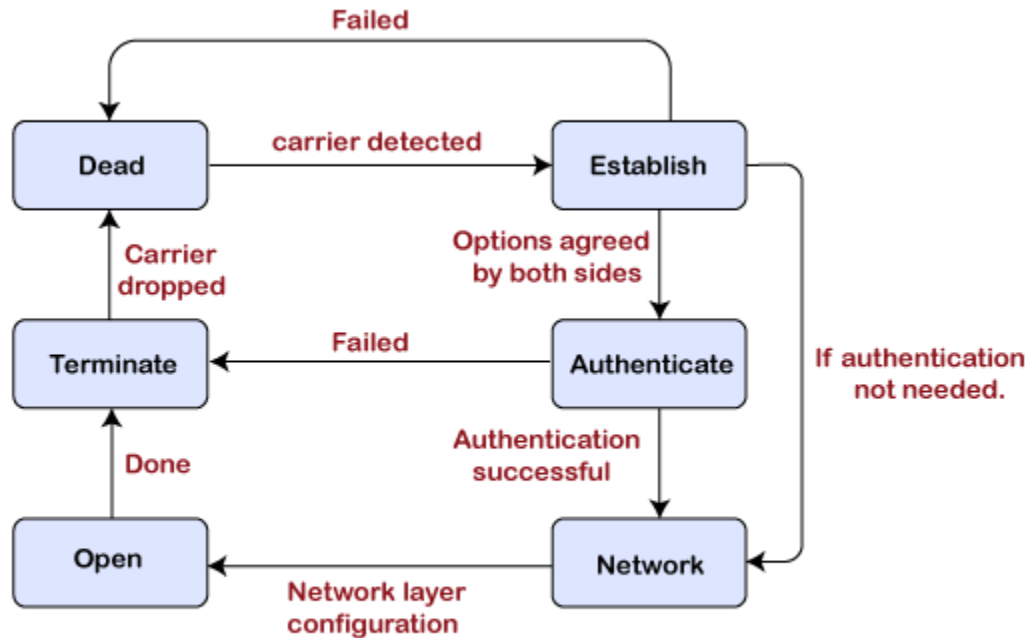


- **Flag:** The flag field is used to indicate the start and end of the frame. The flag field is a 1-byte field that appears at the beginning and the ending of the frame. The pattern of the flag is similar to the bit pattern in HDLC, i.e., 01111110.
- **Address:** It is a 1-byte field that contains the constant value which is 11111111. These 8 ones represent a broadcast message.
- **Control:** It is a 1-byte field which is set through the constant value, i.e., 11000000. It is not a required field as PPP does not support the flow control and a very limited error control mechanism. The control field is a mandatory field where protocol supports flow and error control mechanism.
- **Protocol:** It is a 1 or 2 bytes field that defines what is to be carried in the data field. The data can be a user data or other information.

- **Payload:** The payload field carries either user data or other information. The maximum length of the payload field is 1500 bytes.
- **Checksum:** It is a 16-bit field which is generally used for error detection.

Transition phases of PPP protocol

The following are the transition phases of a PPP protocol:



Transition phases

- **Dead:** Dead is a transition phase which means that the link is not used or there is no active carrier at the physical layer.
- **Establish:** If one of the nodes starts working then the phase goes to the establish phase. In short, we can say that when the node starts communication or carrier is detected then it moves from the dead to the establish phase.
- **Authenticate:** It is an optional phase which means that the communication can also moves to the authenticate phase. The phase moves from the establish to the authenticate phase only when both the communicating nodes agree to make the communication authenticated.
- **Network:** Once the authentication is successful, the network is established or phase is network. In this phase, the negotiation of network layer protocols take place.

- **Open:** After the establishment of the network phase, it moves to the open phase. Here open phase means that the exchange of data takes place. Or we can say that it reaches to the open phase after the configuration of the network layer.
- **Terminate:** When all the work is done then the connection gets terminated, and it moves to the terminate phase.

On reaching the terminate phase, the link moves to the dead phase which indicates that the carrier is dropped which was earlier created.

There are two more possibilities that can exist in the transition phase:

- The link moves from the authenticate to the terminate phase when the authentication is failed.
- The link can also move from the establish to the dead state when the carrier is failed.

PPP Stack

In PPP stack, there are three set of protocols:

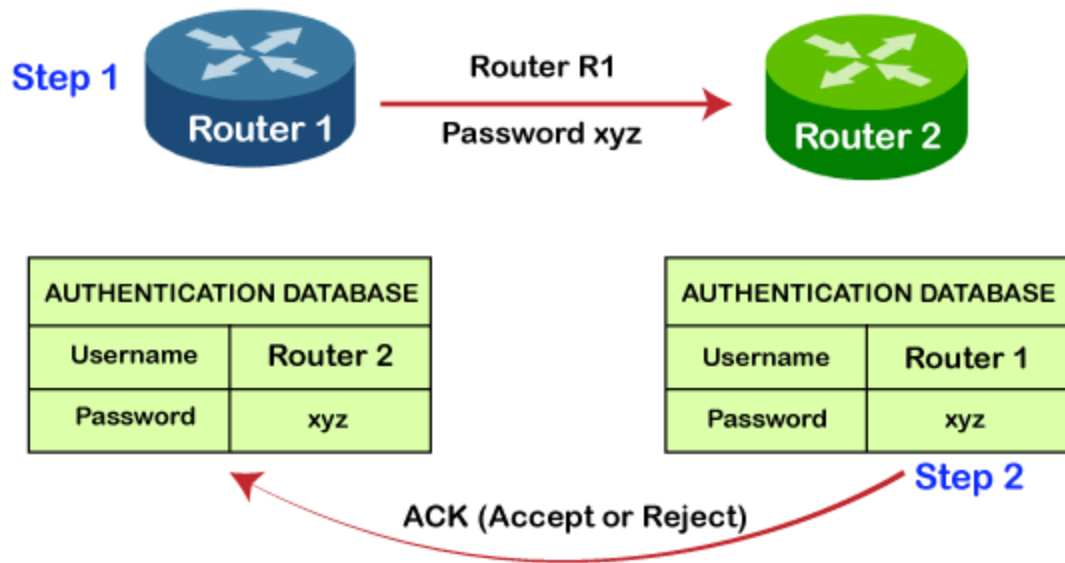
- **Link Control Protocol (LCP)**

The role of LCP is to establish, maintain, configure, and terminate the links. It also provides negotiation mechanism.

- **Authentication protocols**

There are two types of authentication protocols, i.e., PAP (Password Authenticate protocols), and CHAP (Challenged Handshake Authentication Protocols).

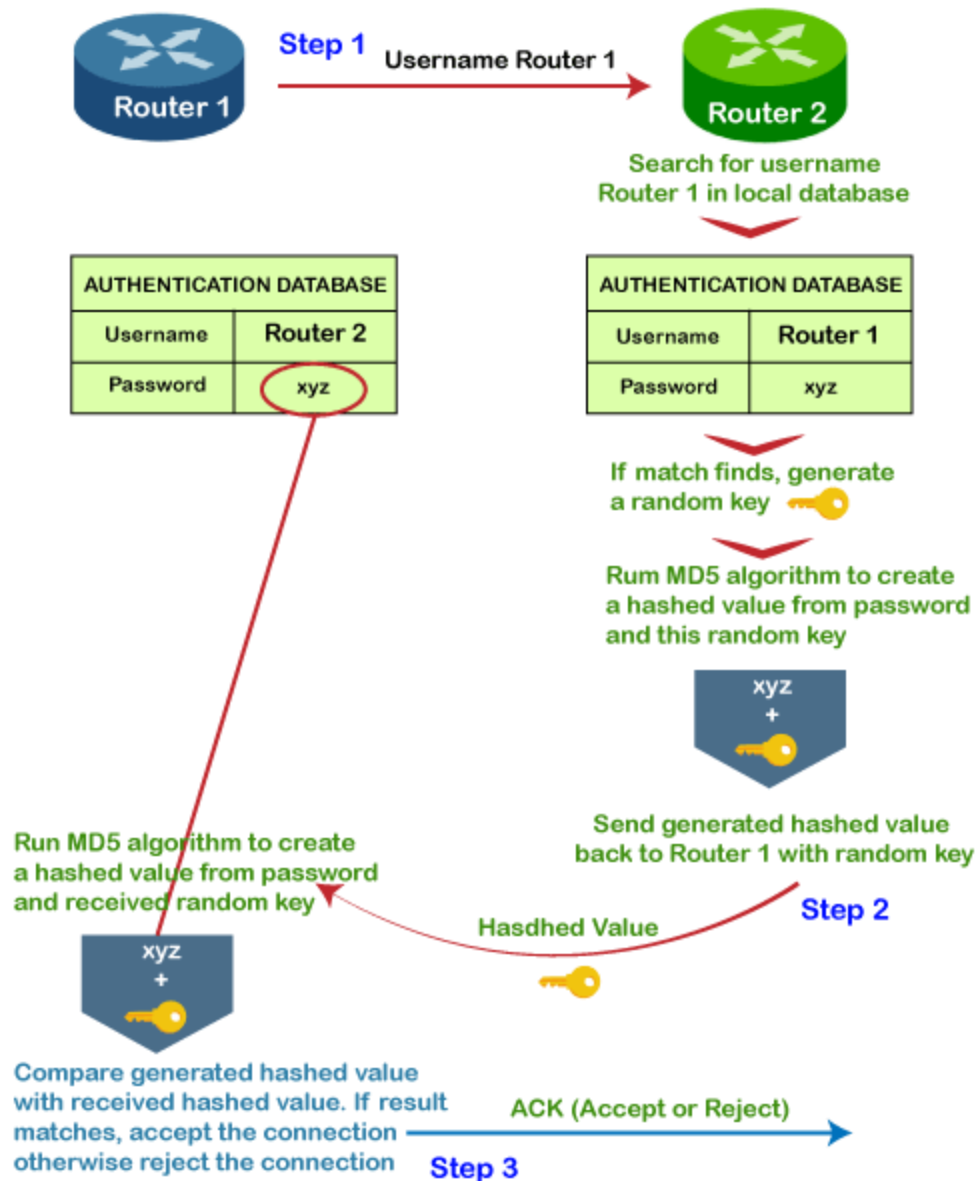
1. PAP (Password Authentication Protocols)



PAP is less secure as compared to CHAP as in case of PAP protocol, password is sent in the form of a clear text. It is a two-step process. Suppose there are two routers, i.e., router 1 and router 2. In the first step, the router 1 wants to authenticate so it sends the username and password for the authentication. In the second step, if the username and password are matched then the router 2 will authenticate the router 1 otherwise the authentication failed.

2. CHAP (Challenged Handshake Authentication Protocol)

CHAP is a three-step process. Let's understand the three steps of CHAP.



Step 1: Suppose there are two routers, i.e., router 1 and router 2. In this step, router 1 sends the username but not the password to the router 2.

Step 2: The router 2 maintains a database that contains a list of allowed hosts with their login credentials. If no data is found which means that the router 1 is not a valid host to connect with it and the connection gets terminated. If the match is found then the random key is passed. This random key along with the password is passed in the MD5 hashing function, and the hashing function generates the hashed value from the password and the random key (password + random key). The hashed value is also known as Challenge. The challenge along with the random key will be sent to the router 1.

Step 3: The router 1 receives the hashed value and a random key from the router 2. Then, the router 1 will pass the random key and locally stored password to the MD5 hashing function. The MD5 hashing function generates the hashed value from the combination of random key and password. If the generated hashed value does not match with the received hashed value then the connection gets terminated. If it is matched, then the connection is granted. Based on the above authentication result, the authentication signal that could be either accepted or rejected is sent to the router 2.

- **Network Control Protocol (NCP)**

After the establishment of the link and authentication, the next step is to connect to the network layer. So, PPP uses another protocol known as network control protocol (NCP). The NCP is a set of protocols that facilitates the encapsulation of data which is coming from the network layer to the PPP frames.